

In-Class Activity 9. Does the model work for everyone?

DSDA 310

Your model has one F1 score. The people it scores do not. Today you split that number by sex and see whether the model works as well for everyone.

Everyone runs the same model today, so every screen shows the same numbers and we can argue about the same thing. Your own Project 1 model is not replaced; you repeat this on your own features afterwards.

What you write today is the evaluation and failure analysis due before Tuesday Oct 6.

Run this first

5 minutes.

Download `activity9_model.py` and run it, or paste the cells. It fits two models and takes about fifteen seconds.

```
NUM =
['age', 'education_num', 'capital_gain', 'capital_loss', 'hours_per_week']

# MODEL A: everything, including sex and race
CAT_A = ['workclass', 'marital_status', 'occupation', 'race', 'sex']

# MODEL B: identical, sex and race removed
CAT_B = ['workclass', 'marital_status', 'occupation']

train = df.fnlwgt % 10 < 8          # the split from Activity 6
RandomForestClassifier(n_estimators=300, min_samples_leaf=5,
random_state=0)
```

Part A. Model A, split by sex

8 minutes.

Model A has sex in it. Fill in what it printed.

Group	n	Base rate	Selected	Precision	Recall
Male	4251	0.310	0.252	0.771	0.628
Female	2117	0.094	0.054	0.800	0.460

A1. Which number differs most between the two rows? Write what it means as a sentence about people, not about metrics.

There are almost double the number of males than females, this could mean that women are under represented in this dataset.

A2. Precision is **higher** for women, 0.800 against 0.771. That sounds like the model treats women better. Does it? Look at the **Selected** column before you answer.

The selected metric for men is 0.252, for women it's 0.054, this means that men are selected significantly more than women, meaning that women can still be underrepresented and the model hasn't converged to an accurate real precision yet.

Part B. Model B, with sex removed

8 minutes.

Model B is the same model with sex and race deleted from the feature list. Before you look: write down what you expect to happen to the recall gap.

Your prediction: The recall will stay the same

Group	n	Base rate	Selected	Precision	Recall
Male	4251	0.310	.0259	0.760	0.636
Female	2117	0.094	0.055	0.829	0.485

B1. Write the recall gap for each model. Did removing sex close it?

0.168 and 0.151, it decreased by a somewhat significant amount

B2. The model can no longer see sex at all. So how is it still treating the two groups differently? Name the feature doing the work.

Features like relationship and marital status can be indicative of sex and native country can be indicative of race.

Part C. So what do you do

10 minutes.

C1. Men are selected at 0.259 and women at 0.055. Should those rates be equal? The base rates in this data are 31.0% and 9.4%. Argue one side in three sentences.

No, the outcome should be representative of the raw data. To make the rates equal you should first clean the data so that there is a 50/50 split in men and women. Selected men and women at intentionally different rates would not be proper random sampling.

C2. You could lower the threshold for women until recall matches. What would that cost, and who pays?

The precision gap would widen.

C3. You are asked to deploy this tomorrow for a decision that affects real people. Yes or no, and name the **one number** you would watch weekly, with the value that would make you stop the model.

I would watch the recall gap until it reaches below 0.05

C4. A bank uses this model to decide who gets an unsolicited premium offer. Only the people it flags get products, so only they generate records, and those records become next year's training data. Model B flags **5.5% of women** and 25.9% of men. What happens to those numbers after three cycles, and what would you do about it?

Women will consistently get less premium offers regardless of if it is better to raise the percentage of women that get the offers.

One idea to leave with

You cannot fix a fairness problem by deleting a column. The information finds another way in. The work is to measure what the model does to each group, choose which definition of fair your use case needs, and report the gap you accepted.

What to hand in

Commit `analysis/activity9_fairness.md` with both tables, A1 and A2, B1 and B2, and C1 to C4. Then repeat Part A on **your own** Project 1 model and add that table.

This is the evaluation and failure analysis milestone, due before class on Tuesday Oct 6.

Logistic Regression

Without Sex/Race:

Train Accuracy: 0.848

Train Precision: 0.739

Train Recall: 0.603

Train F1-score: 0.664

Test Accuracy: 0.846

Test Precision: 0.735

Test Recall: 0.601

Test F1-score: 0.661

With:

Train Accuracy: 0.849

Train Precision: 0.739

Train Recall: 0.608

Train F1-score: 0.667

Test Accuracy: 0.847

Test Precision: 0.734

Test Recall: 0.606

Test F1-score: 0.664

Everything is within 0.01 of its counterpart

Tree

Without

Accuracy: 0.914

Precision: 0.904

Recall: 0.734

F1-score: 0.810

Test Accuracy: 0.859

Test Precision: 0.786

Test Recall: 0.596

Test F1-score: 0.678

With

Accuracy: 0.914

Precision: 0.901

Recall: 0.735

F1-score: 0.810

Test Accuracy: 0.858

Test Precision: 0.784

Test Recall: 0.597

Test F1-score: 0.678

Same thing here, everything is within 0.01 of its counterpart